Tutorial 1.1
Revision on Procedural Programming

Overview

- In this tutorial you will be revising some topics from PT1.
- Also, you will be learning how to get started with Microsot Visual Studio Code for C++ programming.
- You will be doing online collaborative coding to solve the problem given in this activity.

Breakout Sessions

- I will split the main meeting room on Webex into several rooms. Each room will have 4 members.
- You will be assigned to a room. See the attached file to find which room you belong to.
- Do introduce yourself to your room members before getting started.

Collaborative Coding

- In each room, appoint one member to be the host. The host member will initiate a session for collaborative coding using Live Share.
- Other members will be invited as collaborators.
- All members (host and collaborators) will need to open VS Code.
- The host member will also need to share his/her screen via Webex.
- You will need to present your code later.

Problems

Write a program in C++ language for each question below. Each question is provided with a program template. Use the program file as the basis to answer the question.

Program 1: Basic Arithmetics

Write a program that performs basic arithmetic operations for additions (+), subtractions (-), multiplications (*) and divisions (/). Complete the **program1.cpp** for this question.

An example run of the expected program. Note that the bold text indicates user inputs:

```
Enter arithmetic operation => 5  /  2

Result = 2.5
```

Program 2: Find the largest number

Given a function named **`largest`** to determine the largest value of two numbers. Complete the **program2.cpp** to find the largest value of four numbers entered by the user.

Expected run and output.

```
Enter four numbers => 10  4  99  15

The largest value is 99
```

Program 3a: Find the sum of a list of numbers

Complete the **program3a.cpp** to calculate the sum of numbers from a starting number, *m* to an ending number, *n*. The values of *m* and *n* are specified by the user. There are two situations, i.e., $m \geq n$ or $m \leq n$. Your program must be able to handle both situations.

Expected run and output.

Run 1

```
Enter the starting number, m => 1
Enter the ending number, n => 3

The sum is 6
```

Run 2

```
Enter the starting number, m => 4
Enter the ending number, n => 1

The sum is 10
```

Program 3b: Calculate statistics of data

Complete the **program3b.cpp** to determine some descriptive statistics of data entered by the user. The statistics to be calculated are as follows:

- The largest number
- The sum of all numbers
- The mean of the numbers

The program should fulfill the following requirements:

- Store the list of numbers entered by the user into **an array**.
- Define a function for each of the statistics above.

Expected run and output.

Run 1

```
How many numbers do you want to enter?  => 3
Enter number #1  => 1.2
Enter number #2  => 2.8
Enter number #3  => 2

Largest :  2.8
Sum     :  6
Mean    :  2
```

Run 2

```
How many numbers do you want to enter?  => 4
Enter number #1  => 1
Enter number #2  => 2
Enter number #3  => 3
Enter number #4  => 4

Largest :  4
Sum     :  10
Mean    :  2.5
```

Program 4: Date conversion 1

Complete the **program4.cpp** which firstly ask the user to enter his or her date of birth in the format of **dd mm yy** (e.g., 12 9 18) and then convert the date to the format of **Mmm dd, yyyy** (e.g., Sep 12, 2018)

Expected run and output.

```
Enter your DOB (dd mm yy)  => 12  9  18

Your Date of Birth is Sep 12, 2018
```

Program 5: Date conversion 2

Complete the **program5.cpp** which firstly ask the user to enter his or her date of birth in the format of **Mmm dd yyyy** (e.g., Sep 12 2018) and then convert the date to the format of **dd-mm-yyyy** (e.g., 12 9 18)

Expected run and output.

```
Enter your DOB (Mmm dd yyyy)  => Sep 12 2018

Your Date of Birth is 12-9-2018
```

Program 6: Find the covariance of two lists of numbers

Covariance is among statistical measures to describe relatedness between two lists of values. It is expressed by the following equation:

$$covar = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{n}$$

where x and y are the lists of numbers. Thus, x_i is an element (a number) from the list at indexed i, $\bar{x}$ and $\bar{y}$ are the average of each list, respectively, and n the size of each list.

Complete the **program6.cpp** to calculate the covariance value of two arrays of numbers read from input files. An helper function named `readArrayFromFile` is given to read a list of numbers from an input file. Your are also provided with some samples of input files, **list1.txt**, **list2.txt**, and **list3.txt,** to test your program.

Expected run and output:

Run 1

```
Enter file #1  => list1.txt
Enter file #2  => list2.txt

Covariance = -90.96
```

Run 2

```
Enter file #1  => list2.txt
Enter file #2  => list3.txt

Covariance = -155.32
```

Run 3

```
Enter file #1  => list3.txt
Enter file #2  => list1.txt

Covariance = 376.12
```